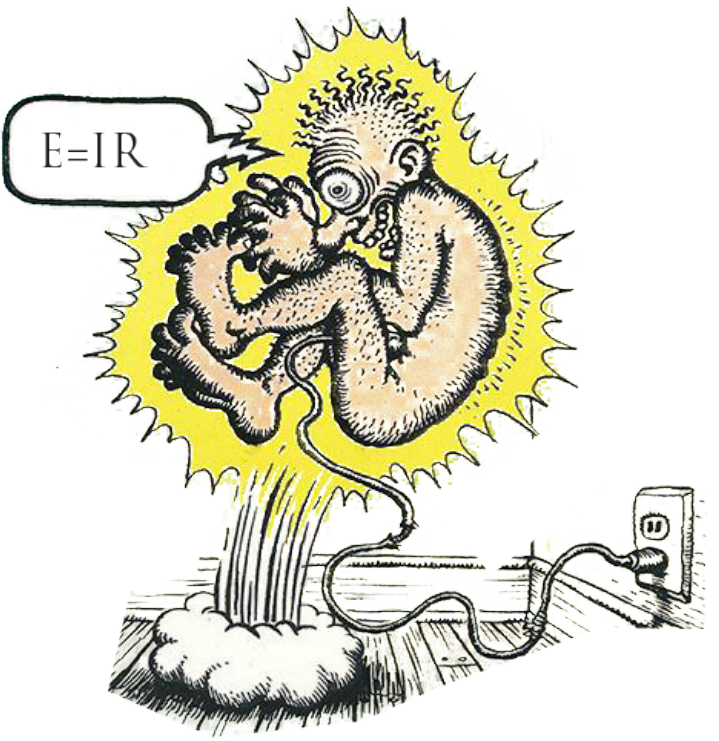
Troubleshoot Out the Lights



Electrician Apprentice
WORKBOOK

ALLIED PRINTING TRADES UNION LABEL COUNCIL NEW YORK

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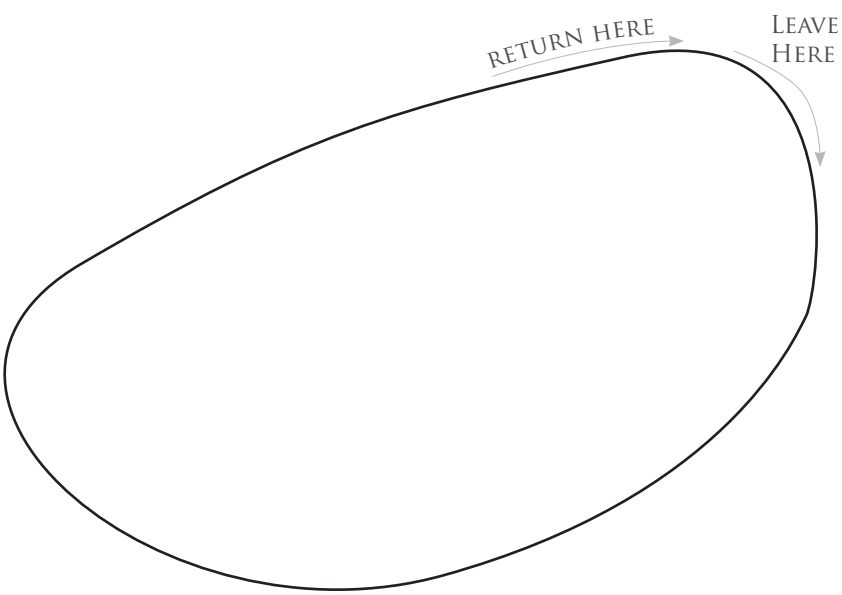
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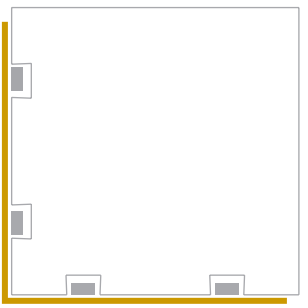
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Chapter 1:
Circuit



The Bent Wire

Grab four circuit tiles like the one to the right. Connect them into a 2 x 2 tile circuit.



When doing so, notice that two sides of each tile have nothing on them, and the other two have both magnets and conductive tape attached. The magnets are there so the tapes from both tiles will press into each other, making an electrical connection.

Imagine you're on an electron moving about the conductive tapes on some of the edges of the 2 x 2 circuit tile grid you made.

Neodyne magnet

Conductive tape

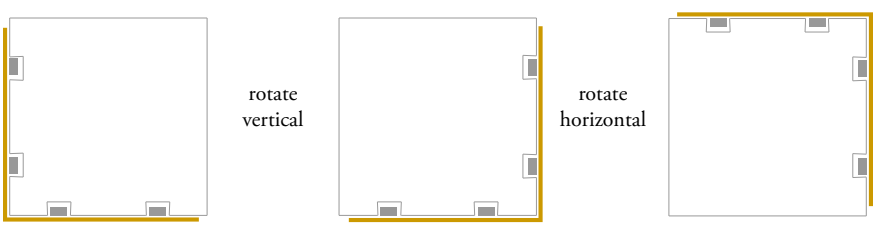
In the tile above, if your electron enters on the left side, where does it leave ?

In one side, out the other

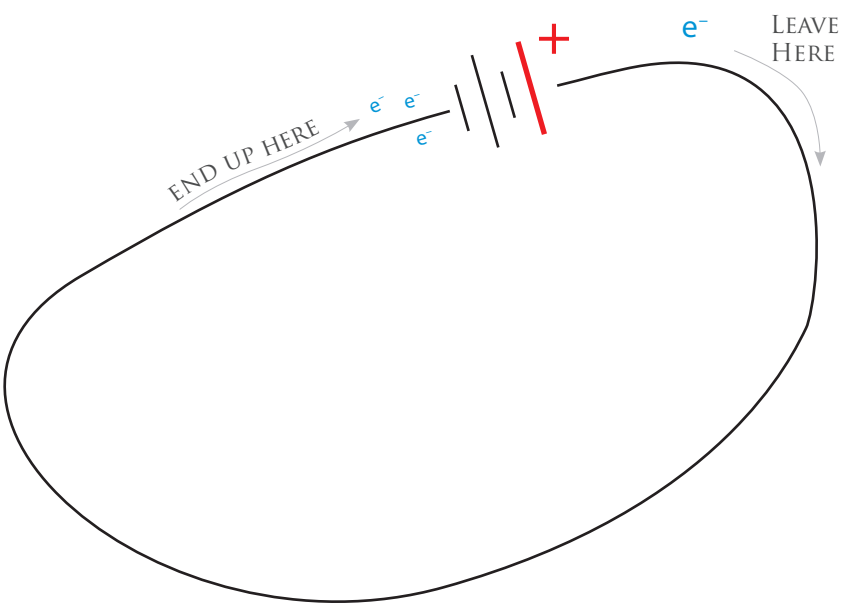
So if your electron enters on one side of a bent wire tire, it has only one other side to exit. There's another limitation, the magnets will only let you connect the tiles in some ways, not others. But you can manipulate the tiles to get around these limitations.

Connect two bent wire tiles, then disconnect them, flip one over, and reconnect the tiles. How will the movement of electrons change?

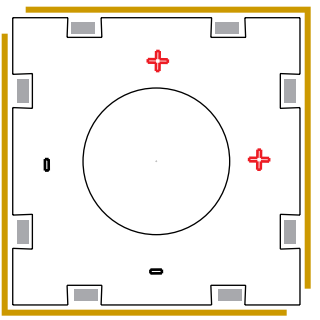
Here are all the manipulations you have with a bent wire tile:



Chapter 2:
Polarity



The Battery Tile



Remove one of the bent wiire circuit tiles in the 2 x 2 grid. Replace it with a battery tile.

Observe how the battery tile will only connect to the circuit grid in some ways, not others. Can you think of any other ways the battery tile will limit how your electron will move along the conductive tapes?

Here's another way to look at it. Get a multimeter, turn it on, and touch the red probe to the front side of the battery, the other to the reverse side. What do you see on the multimeter screen?

+

We saw red this chapter. It's hot, dangerous, where you can get zapped.

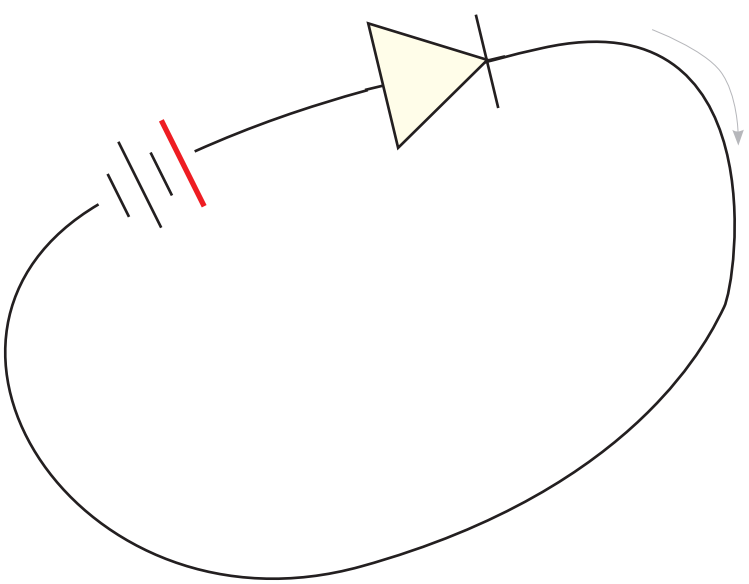
With circuit tiles and conductive tape, no worries.

Touching the multimeter's probes to the battery registers how much electrons want to move closer to the anode, where electrons (e^-) congregate.

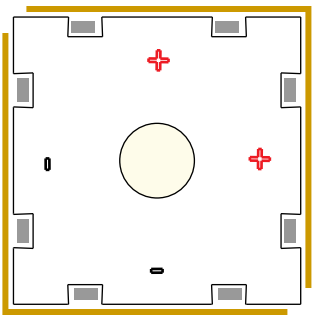
Look at the battery tile again. The number on the voltmeter is how much the electrons on the upper right of the battery tile want to get over to the bottom left of the tile. But they can't do it because they can't travel through wood.

So what do you think the battery tile can do for the rest of the circuit tile grid?

Chapter 3:
LED



The LED Tile



Let's use the battery to light a LED.

Observe how the battery tile will only connect to the circuit grid in some ways, not others. Can you think of any other ways the battery tile will limit how your electron will move along the conductive tapes?

Here's another way to look at it. Get a multimeter, turn it on, and touch the red probe to the front side of the battery, the other to the reverse side. What do you see on the multimeter screen?

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We saw red this chapter. It's hot, dangerous, where you can get zapped.

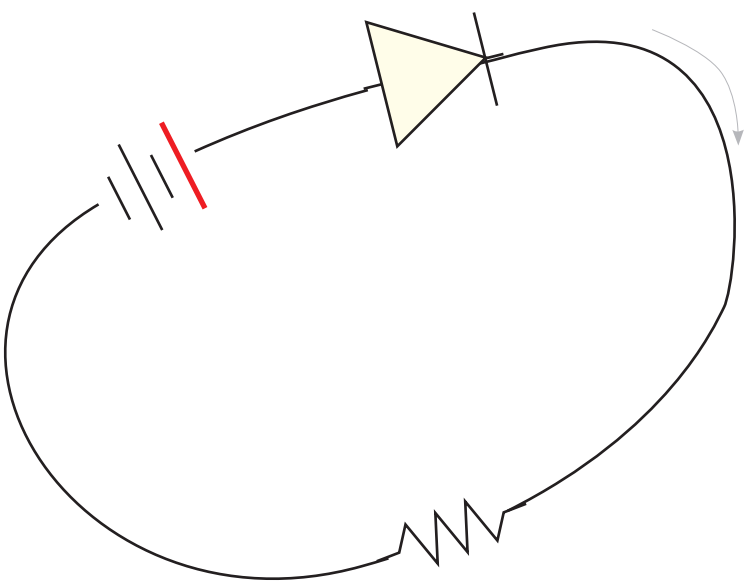
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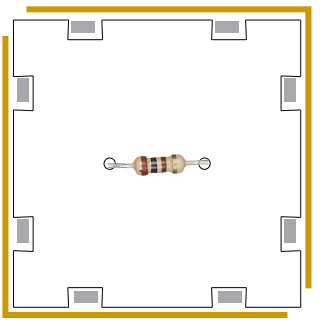
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So what do you think the battery tile can do for the rest of the circuit tile grid?

Chapter 4:
Ohm's Law



The Resistor Tile



Was the LED a bit too bright, burning your eyes? The resistor tile can reduce how brightly the LED emits light.

Pull serveral resistor tiles with different values on them. Notice the different band colors for the different resistor values (see Appendix)

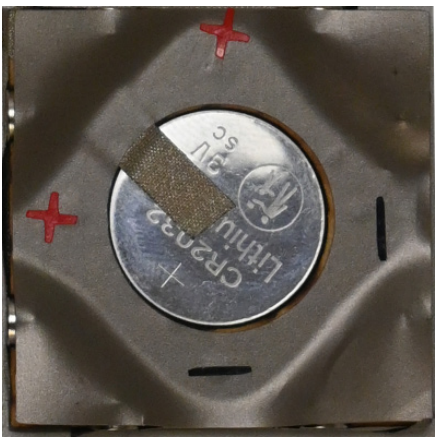
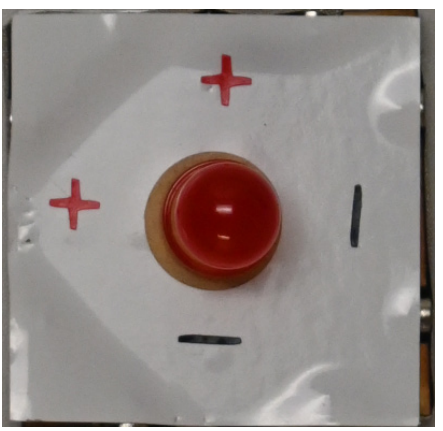
List the different resistor values (in ohms) down the left column, and in the right describe how the LED changes.

Resistor Value	LED light quality

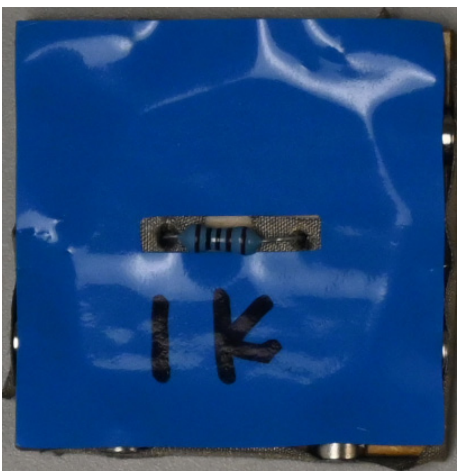
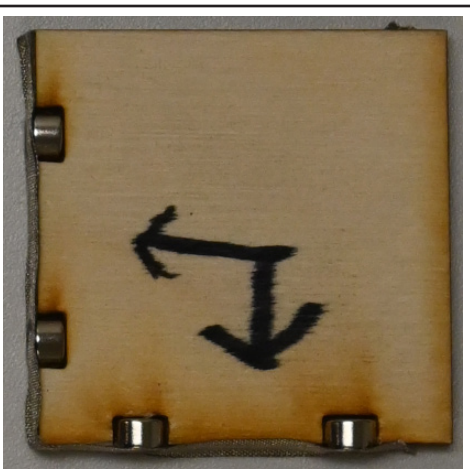
What is the relationship between resistance and current?

Chapter 7: Appendix

Circuit Tiles

Tile	Picture
Battery	
LED	

Circuit Tiles (cont'd)

Resistor	
Bent Wire	

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Circuit Tiles (cont'd)

Straight Wire	
Wire Branch	

Significant Figures
Multiplier
Tolerance
0 Black
1 Brown
2 Red
3 Orange
4 Yellow
5 Green
6 Blue
7 Violet
8 Gray
9 White
x 10⁻³ Silver
x 10⁻¹ Gold
x 10⁰ Black
x 10¹ Brown
x 10² Red
x 10³ Orange
x 10⁴ Yellow
x 10⁵ Green
x 10⁶ Blue
x 10⁷ Violet
x 10⁸ Gray
x 10⁹ White
±20% White
±10% Silver
±5% Gold
±1% Brown
±2% Red
±0.5% Green
±0.25%Blue
±0.1% Violet
±0.05%Gray
250Ω ±5%
46Ω ±1%
276 kΩ ±0.5%
4 Band
5 Band
6 Band
Tolerance
Multiplier
Significant Figures

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